

Phase 5 – API Design

Chapter 1 – API Overview

1.1 Purpose

The purpose of this document is to define all REST APIs required for the **Gold Trading Automation Platform (GTAP)**. It specifies API endpoints, request and response formats, authentication requirements, error handling, and communication between the frontend, backend, and trading engine.

1.2 API Style

The project follows the **RESTful API** architecture.

Base URL

`http://localhost:5000/api/v1`

Production:

`https://your-domain.com/api/v1`

1.3 Content Type

All APIs will use:

Content-Type: `application/json`

1.4 Authentication

Protected APIs will use **JWT (JSON Web Token)**.

Example Header:

Authorization: Bearer <JWT_TOKEN>

1.5 API Naming Convention

Rules:

- Use nouns instead of verbs.
- Use lowercase URLs.
- Use plural resources where applicable.
- Keep endpoint names meaningful.

Examples:

/api/v1/auth/login

/api/v1/dashboard

/api/v1/trades

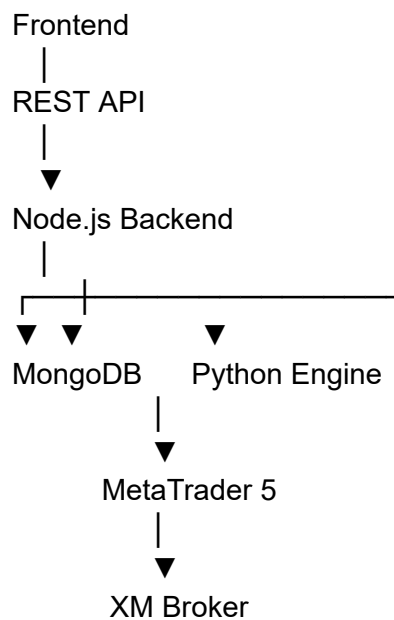
/api/v1/reports

/api/v1/settings



Chapter 2 – API Architecture

API Flow



Responsibilities

Frontend

- Send API Requests

- Display Data
 - Show Errors
 - Receive Live Updates
-

Backend

- Validate Requests
 - Authenticate Users
 - Read/Write Database
 - Call Python Engine
 - Return Response
-

Python Engine

- Receive Commands
 - Execute Strategy
 - Connect MT5
 - Return Trade Result
-

MongoDB

- Store Users
 - Store Trades
 - Store Reports
 - Store Logs
-



Chapter 3 – Authentication APIs

API 1 – Login

Endpoint

POST /api/v1/auth/login

Purpose

Authenticate user and return JWT Token.

Request Body

```
{  
  "email": "user@example.com",  
  "password": "password123"  
}
```

Success Response

```
{  
  "success": true,  
  "message": "Login Successful",  
  "token": "JWT_TOKEN",  
  "user": {  
    "id": "123",  
    "name": "Vinit Kushwaha"  
  }  
}
```

Error Response

```
{  
  "success": false,  
  "message": "Invalid Email or Password"  
}
```

API 2 – Logout

Endpoint

POST /api/v1/auth/logout

Purpose:

Destroy user session.

API 3 – Verify Token

Endpoint

GET /api/v1/auth/verify

Purpose:

Check JWT validity.



Chapter 4 – User APIs

API 1 – Get Profile

GET /api/v1/users/profile

API 2 – Update Profile

PUT /api/v1/users/profile

API 3 – Change Password

PUT /api/v1/users/change-password

अब हम **Phase 5 – API Design** का अगला हिस्सा शुरू करते हैं।



Chapter 5 – Dashboard APIs

Dashboard सभी Real-Time Data दिखाएगा।

API 1 – Get Dashboard Summary

Endpoint

GET /api/v1/dashboard

Authentication

✓ JWT Required

Purpose

Dashboard की मुख्य जानकारी प्राप्त करना।

Success Response

```
{
  "success": true,
  "data": {
    "balance": 10000,
    "equity": 10250,
    "dailyProfit": 250,
    "activeTrades": 2,
    "winRate": 76
  }
}
```

API 2 – Get Trade Statistics

Endpoint

GET /api/v1/dashboard/statistics

Purpose:

- Total Trades
 - Winning Trades
 - Losing Trades
 - Drawdown
 - Net Profit
-

API 3 – Get Active Trades

Endpoint

GET /api/v1/dashboard/active-trades

API 4 – Get Recent Trades

Endpoint

GET /api/v1/dashboard/recent-trades



Chapter 6 – Trading APIs

API 1 – Start Trading Bot

POST /api/v1/trading/start

Purpose

Trading Bot Start करना।

Request

```
{  
  "symbol": "XAUUSD"  
}
```

Response

```
{  
  "success": true,  
  "message": "Trading Bot Started Successfully"  
}
```

API 2 – Stop Trading Bot

POST /api/v1/trading/stop

API 3 – Bot Status

GET /api/v1/trading/status

Response

```
{  
  "status": "Running"  
}
```

API 4 – Get Trade History

GET /api/v1/trading/history

API 5 – Get Trade Details

GET /api/v1/trading/:tradeId



Chapter 7 – Reports APIs

API 1 – Daily Report

GET /api/v1/reports/daily

API 2 – Weekly Report

GET /api/v1/reports/weekly

API 3 – Monthly Report

GET /api/v1/reports/monthly

API 4 – Custom Date Report

GET /api/v1/reports/custom

Query Parameters

startDate=2026-07-01

endDate=2026-07-31



Chapter 8 – Bot Settings APIs

API 1 – Get Settings

GET /api/v1/settings

API 2 – Update Settings

PUT /api/v1/settings

Request Body

```
{  
  "ema":11,  
  "rsi":14,  
  "lotSize":0.20,  
  "riskReward":"1:2"  
}
```

API 3 – Reset Settings

POST /api/v1/settings/reset



Chapter 9 – Trade Logs APIs

API 1 – Get All Logs

GET /api/v1/logs

API 2 – Get Logs by Trade

GET /api/v1/logs/:tradeId

API 3 – Get Error Logs

GET /api/v1/logs/errors



Chapter 10 – WebSocket Events

Dashboard को **Real-Time** बनाने के लिए हम **Socket.IO** का उपयोग करेंगे।

Event 1 – Bot Started

Event Name

botStarted

Data

```
{  
  "status": "running",  
  "message": "Trading Bot Started"  
}
```

Event 2 – Bot Stopped

botStopped

Event 3 – Trade Executed

tradeExecuted

Data

```
{  
  "symbol": "XAUUSD",  
  "orderType": "BUY",  
  "entryPrice": 3352.40,  
  "lotSize": 0.20  
}
```

Event 4 – Trade Closed

tradeClosed

Event 5 – Dashboard Updated

dashboardUpdated

Event 6 – MT5 Connection Status

mt5Connection

Possible Values:

- Connected
 - Disconnected
 - Reconnecting
-

Chapter 11 – Standard Request Format

सभी APIs एक समान Structure Follow करेंगी।

Example

```
{  
  "email": "user@example.com",  
  "password": "password123"  
}
```

Rules

- JSON Format
 - camelCase Keys
 - UTF-8 Encoding
 - Proper Validation
-

Chapter 12 – Standard Response Format

Success Response

```
{  
  "success": true,  
  "message": "Operation Successful",  
  "data": {}  
}
```

Error Response

```
{  
  "success": false,  
  "message": "Something went wrong",  
  "error": {}  
}
```

Pagination Response

```
{
  "success": true,
  "page": 1,
  "limit": 20,
  "total": 250,
  "data": []
}
```



Chapter 13 – HTTP Status Codes

Code	Meaning
200	Success
201	Created
400	Bad Request
401	Unauthorized
403	Forbidden
404	Not Found
409	Conflict
422	Validation Error
429	Too Many Requests (<i>Future</i>)
500	Internal Server Error



Chapter 14 – Error Handling

Authentication Errors

- Invalid Email
 - Wrong Password
 - JWT Expired
 - JWT Invalid
-

Trading Errors

- MT5 Not Connected
 - Market Closed
 - Insufficient Margin
 - Invalid Lot Size
 - Order Execution Failed
-

Database Errors

- Duplicate Email
 - Database Connection Failed
 - Document Not Found
-

Server Errors

- Internal Server Error
 - Timeout
 - Unknown Exception
-

Validation Errors

- Required Field Missing
 - Invalid Data Type
 - Invalid Symbol
 - Invalid Risk Reward
-



Chapter 15 – API Best Practices

Versioning

/api/v1/

Authentication

- JWT
 - Protected Routes
 - Token Expiry
-

Validation

- Request Validation
 - Response Validation
-

Security

- Password Hashing
 - HTTPS (Production)
 - Environment Variables
 - CORS Configuration
-

Logging

- API Logs
 - Error Logs
 - Request Logs
-

Performance

- Pagination
 - Compression
 - Optimized Queries
-

Documentation

सभी APIs को भविष्य में **Swagger/OpenAPI** से Document किया जाएगा।



API Summary

Module	APIs
Authentication	3
User	3
Dashboard	4
Trading	5
Reports	4
Settings	3
Logs	3

Total Planned REST APIs: लगभग 25

इसके अलावा लगभग **6 WebSocket Events** होंगे।